

Some new faces of Bregman divergences

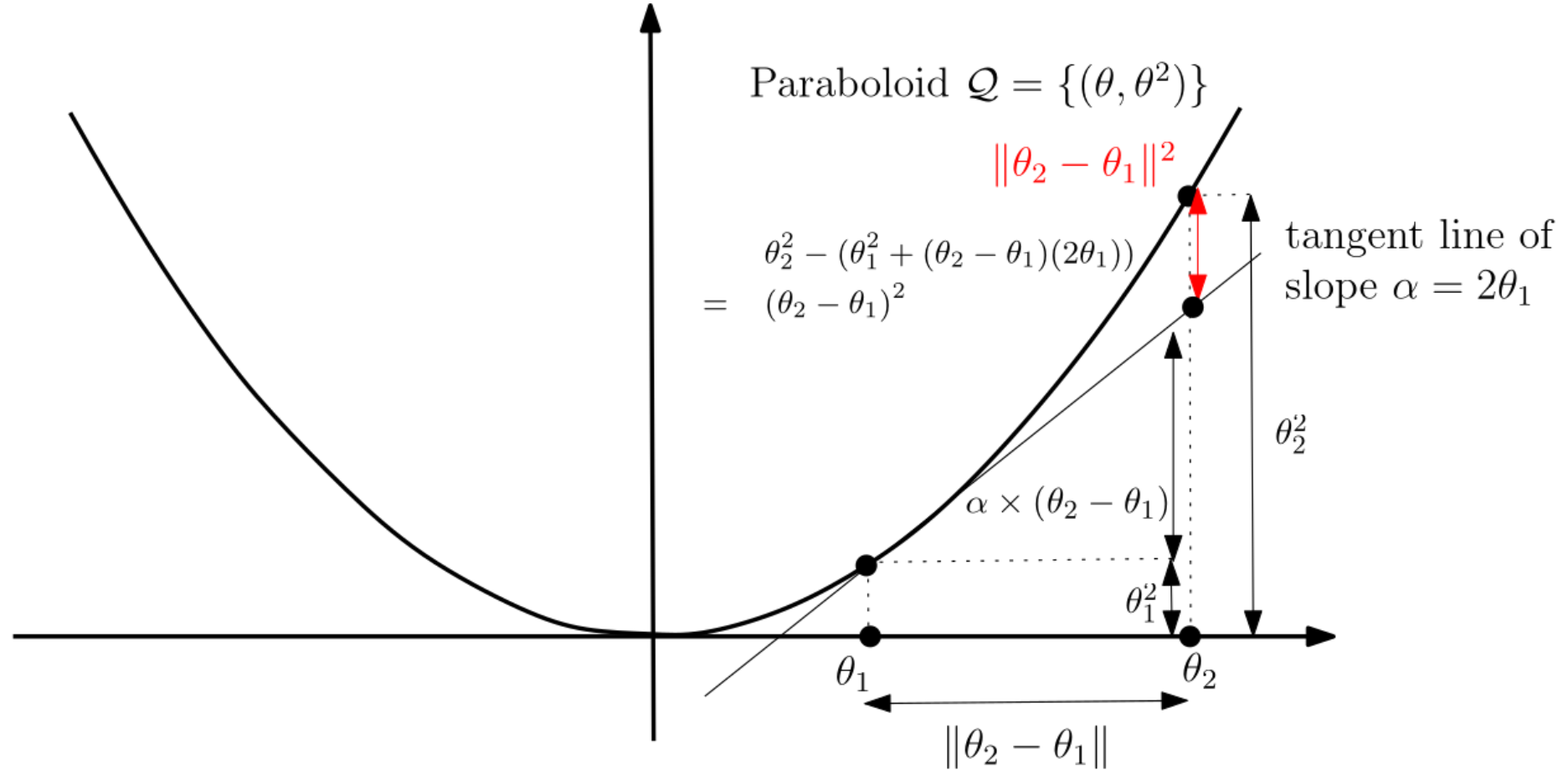
Frank Nielsen

Sony Computer Science Laboratories, Inc.



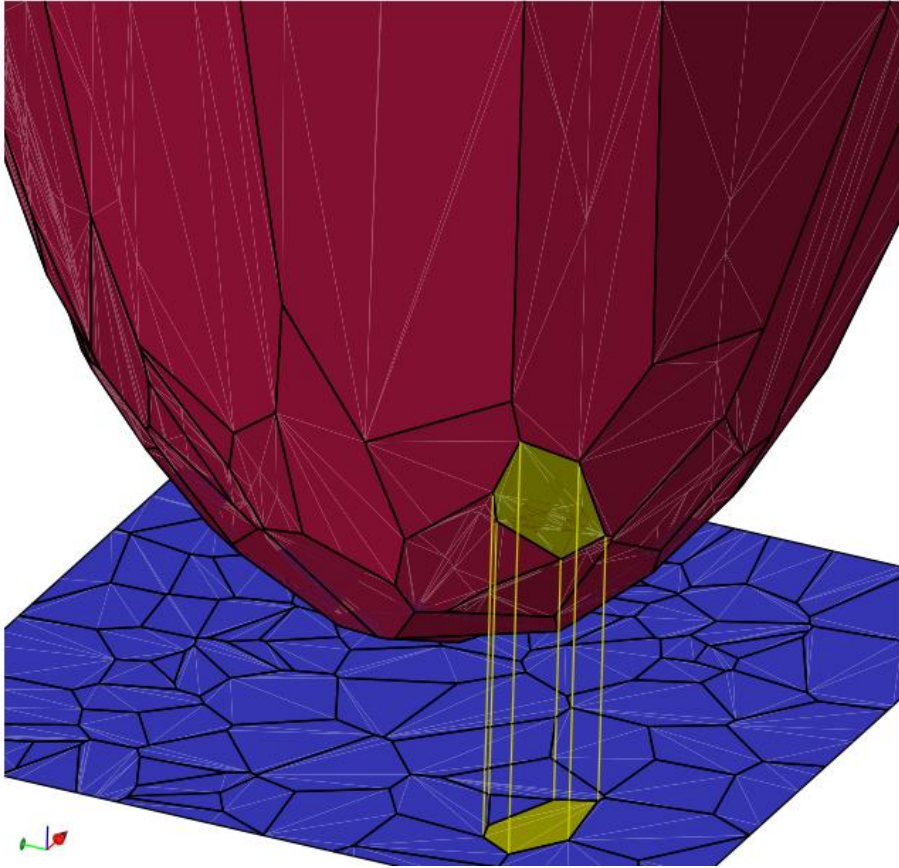
Sony CSL

A graph view of the squared Euclidean distance



- Squared Euclidean distance is **not a metric** because it violates the triangle inequality.
- Smooth distances are called **divergences**

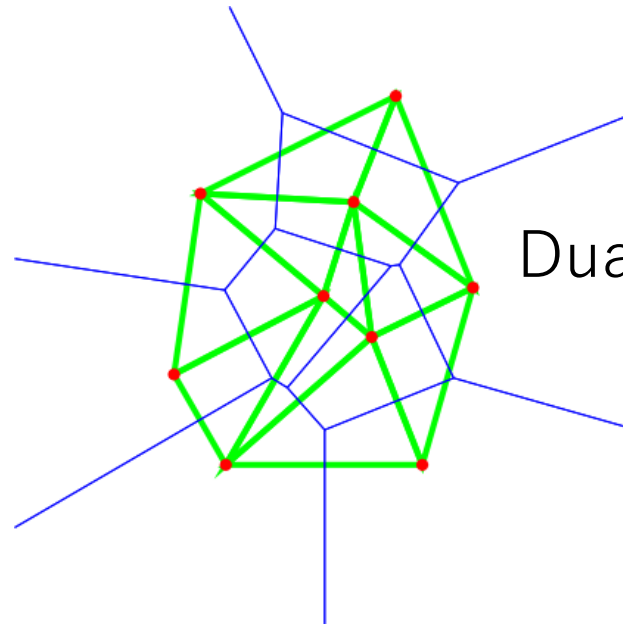
Paraboloid lifting in computational geometry



Illustrations courtesy of J.-D. Boissonnat

Voronoi diagram from intersection of half-spaces after lifting sites to hyperplanes tangent to the paraboloid in higher dimension

$$\mathcal{V}(P) = \bigcap_i h_{p_i}^+$$



Dual **orthogonal** structures

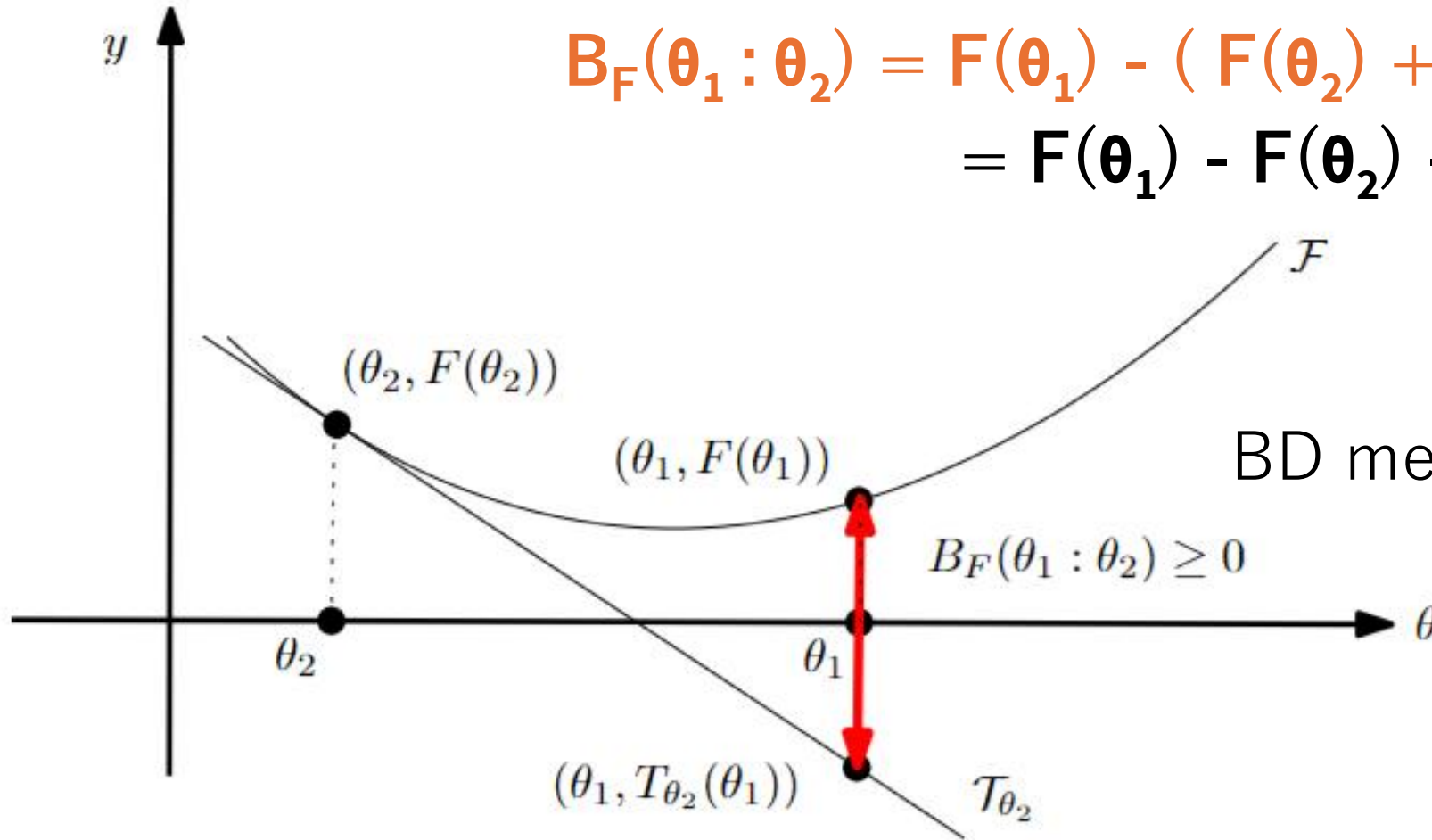
Dual **Delaunay** mesh

Bregman divergences: Convex potential functions

- $F: \Theta \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ a strictly convex and smooth real-valued function on a finite-dimensional Hilbert space equipped with inner product $\langle \cdot, \cdot \rangle$

Bregman divergence $B_F: \Theta \times \text{Int}(\Theta) \rightarrow \mathbb{R}$

$$\begin{aligned} B_F(\theta_1 : \theta_2) &= F(\theta_1) - (F(\theta_2) + \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle) \\ &= F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle \end{aligned}$$



Bregman divergences unify key distortion measures

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle$$

- **Quadratic negentropy** generator yields **squared Euclidean divergence**:

$$F(\theta) = \frac{1}{2} \theta^T Q \theta \text{ with } \Theta = \mathbb{R}^d$$

Computational geometry L_2^2

$$\nabla F(\theta) = Q \theta$$

$$B_F(\theta_1 : \theta_2) = \frac{1}{2} (\theta_2 - \theta_1)^T Q (\theta_2 - \theta_1) = \frac{1}{2} \|\theta_2 - \theta_1\|_Q^2$$

- **Shannon negentropy** generator yields **relative entropy** as known as *Kullback-Leibler divergence*:

$$F(\theta) = \sum_i \theta_i \log(\theta_i) \text{ with } \Theta = \Delta^d \text{ (standard/probability simplex)}$$

$$\nabla F(\theta) = [\log(\theta_i) - \theta_i]_i$$

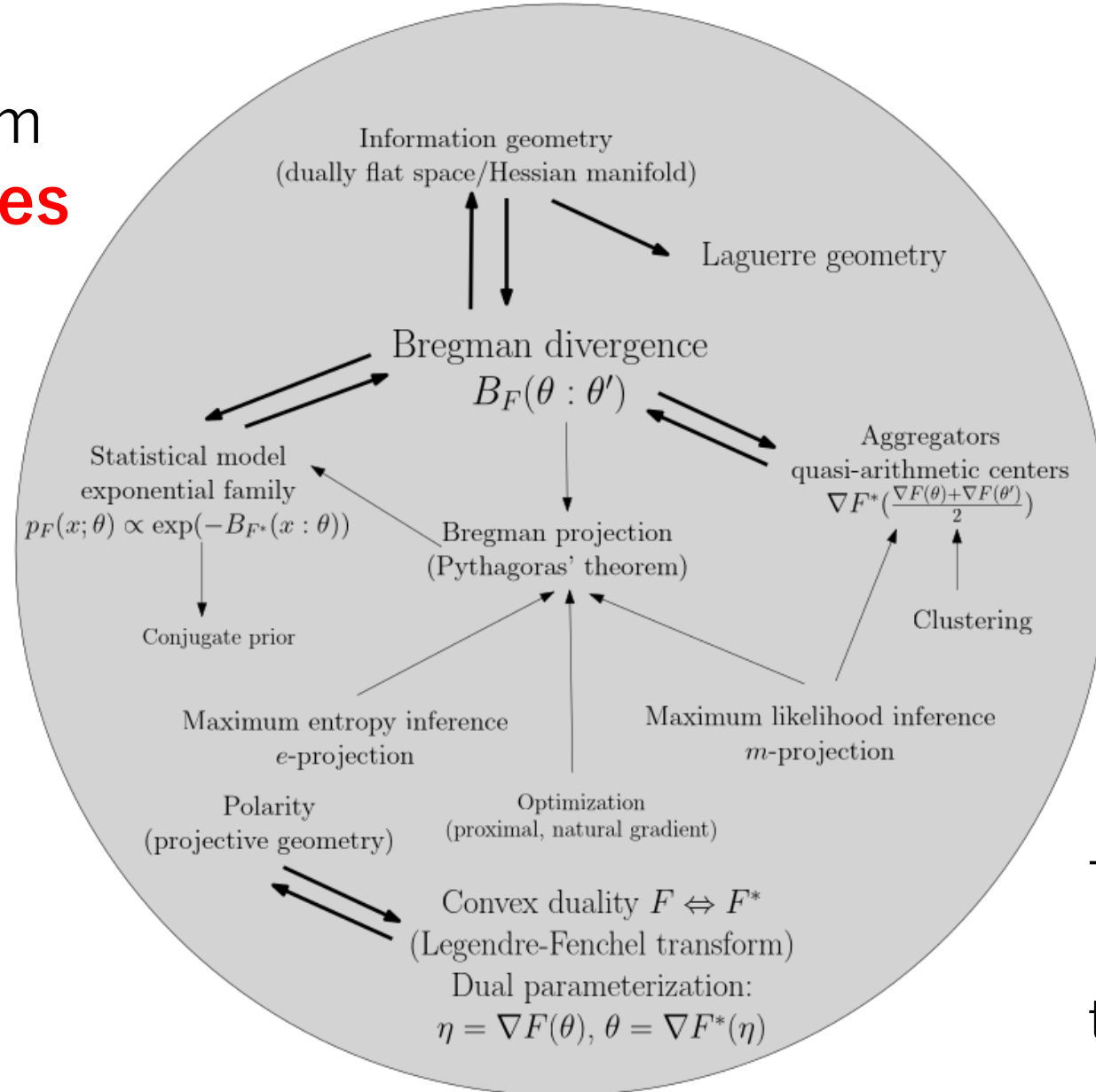
$$B_F(\theta : \theta') = \sum_i \{ \theta_i \log(\theta_i / \theta'_i) + \theta'_i - \theta_i \} = \sum_i \theta_i \log(\theta_i / \theta'_i)$$

$$= \text{KL}(\theta_i : \theta') \text{ since } \sum_i \theta_i = \sum_i \theta'_i = 1$$

Information geometry

Bregman divergences are the workhorses in ML/AI

Ecosystem
of **dualities**

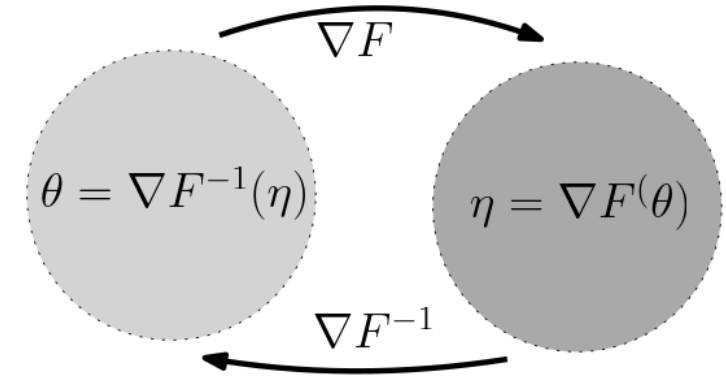


This talk focuses on leveraging
computational geometry
to information geometry

Dual Bregman/Fenchel-Young divergences

- **Gradient map** ∇F yields **dual parameterization**

$$\eta = \nabla F(\theta)$$



- Build the **Fenchel-Young divergence** from Young inequality:

$$F(\theta_1) + F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$$

$$Y_{F, F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$$

$$B_F(\theta_1 : \theta_2) = Y_{F, F^*}(\theta_1 : \eta_2) = Y_{F^*, F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

where F^* is the **Legendre-Fenchel transform** of F :

$$F^*(\eta) = \sup_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - F(\theta) \}$$

$$\text{with } \theta = \nabla F^{-1}(\eta) = \nabla F^*(\eta)$$

Recent new faces of Bregman divergences

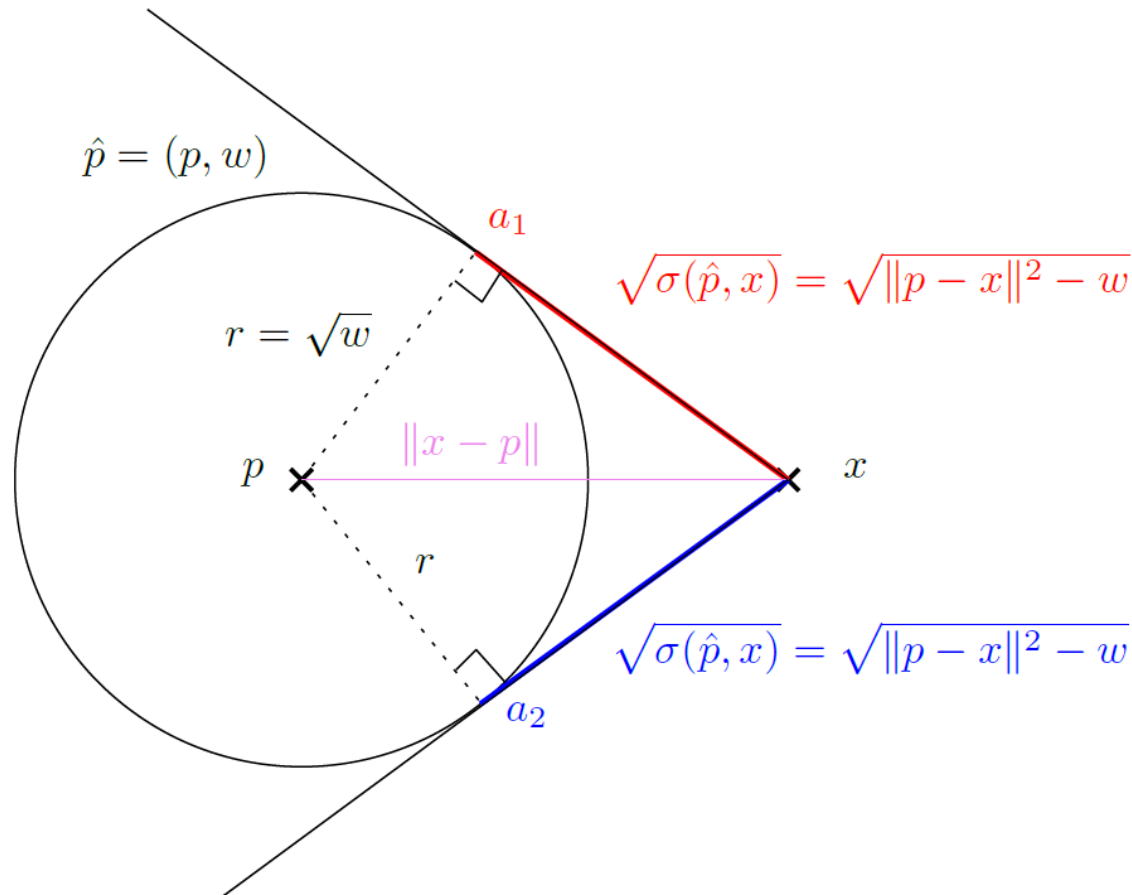
- Bregman Divergences viewed as **power distances** (independent of F)
- Bregman Divergences from **quadratic polarities** (GPU)
- (Functional Bregman divergences: **kernelized functional BDs**)

Power distance between a point and a weighted point

Not a proper distance, can be negative

$$\sigma(x, \hat{p}) = \|x - p\|^2 - w$$

$$\hat{p} = (p, w)$$

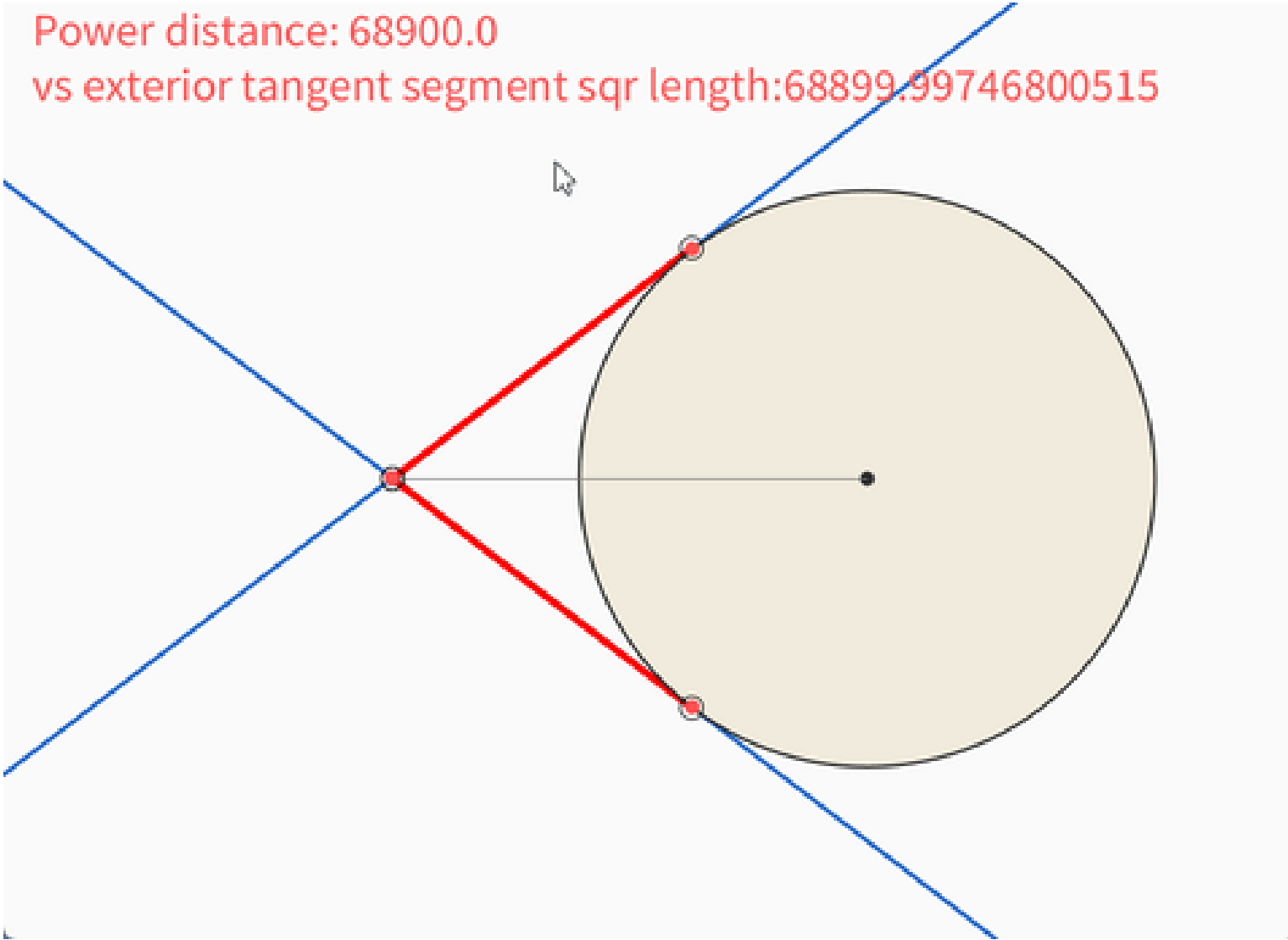


Interpretation follows from the Pythagoras' theorem when positive weight

Between two weighted points: $\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w' = \sigma(\hat{p}', \hat{p})$

Visualizing power distances

Power distance: 68900.0
vs exterior tangent segment sqr length: 68899.99746800515



Power distance is negative when x inside the disk

$$\sigma(x, \hat{p}) = \|x - p\|^2 - w$$

Power distance
studied in
Laguerre geometry

Bregman divergences as power distances

Start with FY divergence $Y_F(\theta : \eta') := F(\theta) + F^*(\eta') - \langle \theta, \eta' \rangle$

and use inner product identity $-\langle \theta, \eta' \rangle = \frac{1}{2} (\|\theta - \eta'\|^2 - \|\theta\|^2 - \|\eta'\|^2)$

to get

$$B_F(\theta : \theta') = Y_F(\theta : \eta') = \frac{1}{2} \sigma(\hat{\theta} : \check{\eta}')$$

Power distance $\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$

$$\hat{p} = (p, w) \quad \hat{p}' = (p', w')$$

Weighted points expressed in **mixed parameterizations**:

$$\hat{\theta} = (\theta, \|\theta\|^2 - 2F(\theta)) \quad \check{\eta} = (\eta, \|\eta\|^2 - 2F^*(\eta))$$

Kullback-Leibler minimax called redundancy

$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\text{KL}}(P : Q)$$

“Minimize maximum error”

Radius of KL miniball

$$D_{\text{KL}}(P : Q) = E_P[\log \frac{1}{Q(X)}] - H(P)$$

$$\mathcal{Q} = M_1^+(\mathcal{X})$$

$$\mathcal{P} = \{P_1, \dots, P_n\} \subset M_1^+(\mathcal{X})$$

Redundancy:
A key measure in
ML/IT

- universal compression (source coding)
- online learning (sequential prediction)
- hypothesis testing
- minimum description length principle
- gambling
- model integration/fusion

Right KL minimax as left Bregman miniballs

$\mathcal{E} = \{P_\theta : \theta \in \Theta\}$ Exponential family with cumulant function F
(Bernoulli/categorical, Gaussian, Wishart, etc.)

KLD between two exponential family densities amounts to a Bregman divergence:

$$D_{\text{KL}}(P_{\theta_i} : P_\theta) = B_F(\theta : \theta_i)$$

When distributions belong to a same exponential family with cumulant function F

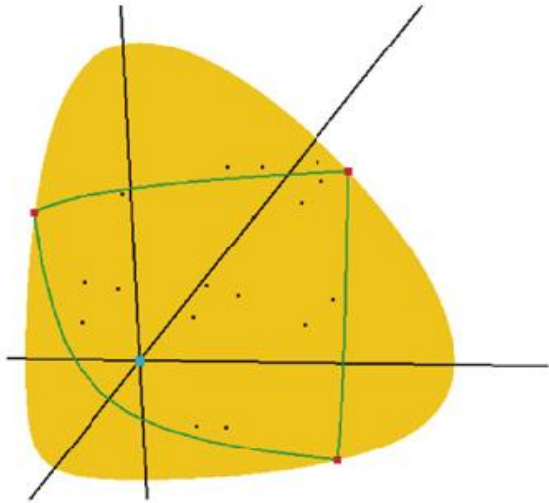
$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\text{KL}}(P : Q) \quad \longrightarrow \quad \min_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

Right-sided KLD minimization

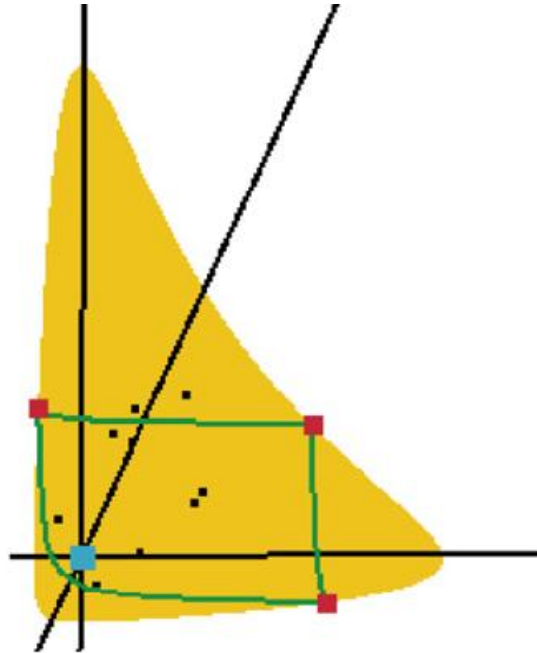
Left-sided BD minimization

Smallest/Minimum enclosing Bregman balls

$$\min_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

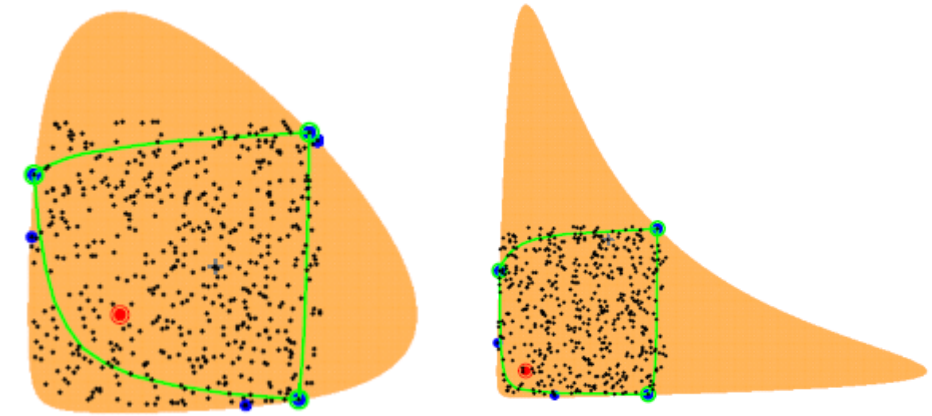


Kullback-Leibler



Itakura-Saito

Linear Programming-type



Algorithm 2: BBC(S)

Input: Data $S = \{s_1, s_2, \dots, s_m\}$;

Output: Center c ;

Choose at random $c \in S$;

for $t = 1, 2, \dots, T - 1$ **do**

$s \leftarrow \arg \max_{s' \in S} D_F(c, s')$;

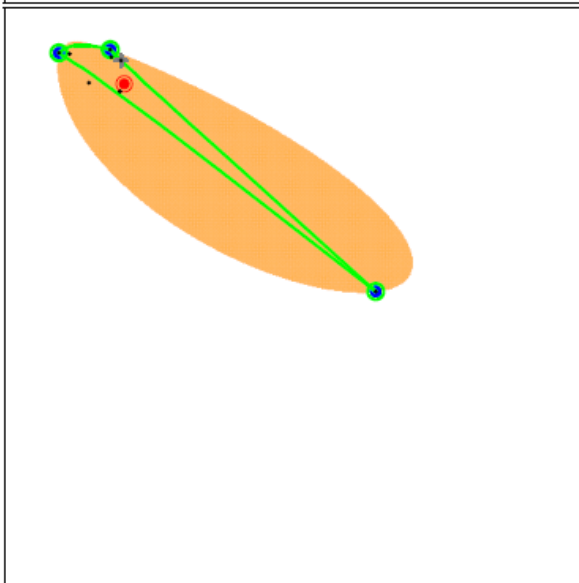
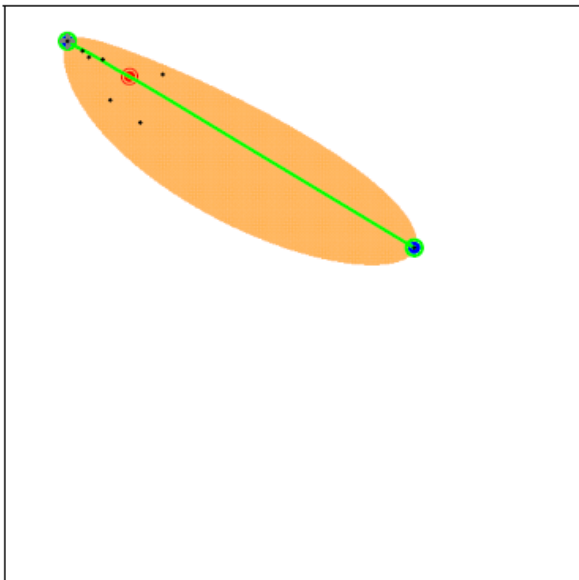
$c \leftarrow \nabla_F^{-1} \left(\frac{t}{t+1} \nabla_F(c) + \frac{1}{t+1} \nabla_F(s) \right)$;

Frank-Wolfe type

"On the smallest enclosing information disk." Information Processing Letters 105.3 (2008)

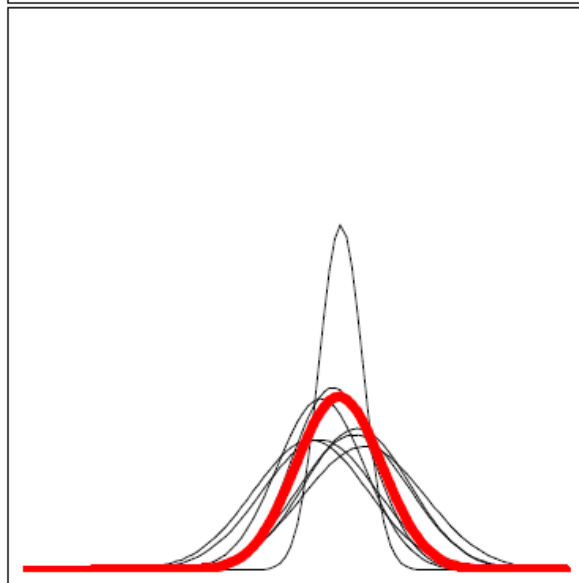
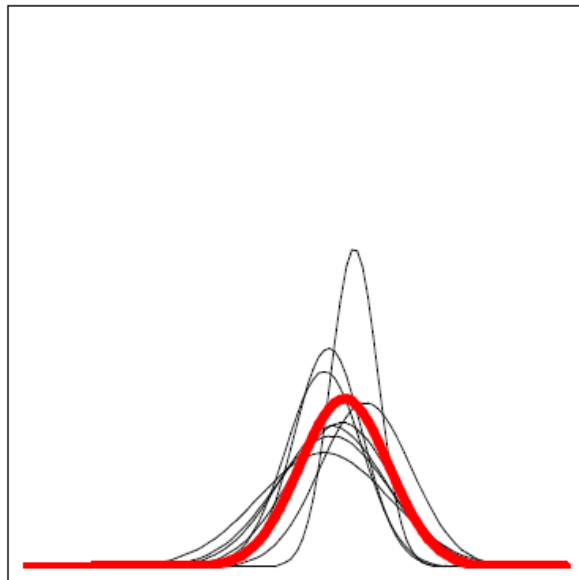
"Fitting the smallest enclosing Bregman ball." ECML 2005.

Natural parameter θ



$$(\theta_1, \theta_2) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right)$$

Gaussian density $N(\mu, \sigma)$



$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x-\mu)^2}{2\sigma^2} \right) \text{ for } x \in \mathbb{R}$$

KL miniball of Gaussians

$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\text{KL}}(P : Q)$$

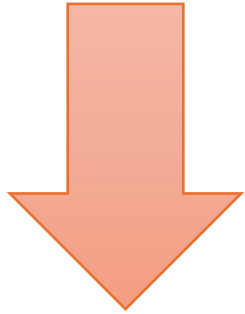
$$\min_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

$$\min_p \max_i \sigma(p : \hat{p}_i)$$

$$F(\theta_1, \theta_2) = -\frac{\theta_1^2}{4\theta_2} + \frac{1}{2} \log \left(-\frac{\pi}{\theta_2} \right)$$

Bregman MEB circumcenter as a power MEB circumcenter

Bregman circumcenter



Power circumcenter

$$\boxed{\theta^* = p^*}$$

Power distance

$$\theta^* = \operatorname{argmin}_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

$$= \operatorname{argmin}_{\theta} \max_i Y_F(\theta : \eta_i)$$

$$= \operatorname{argmin}_{\theta} \max_i \frac{1}{2} \sigma(\hat{\theta} : \check{\eta}_i)$$

$$= \operatorname{argmin}_p \max_i \sigma(p : \hat{p}_i) := p^*$$

Weighted points:

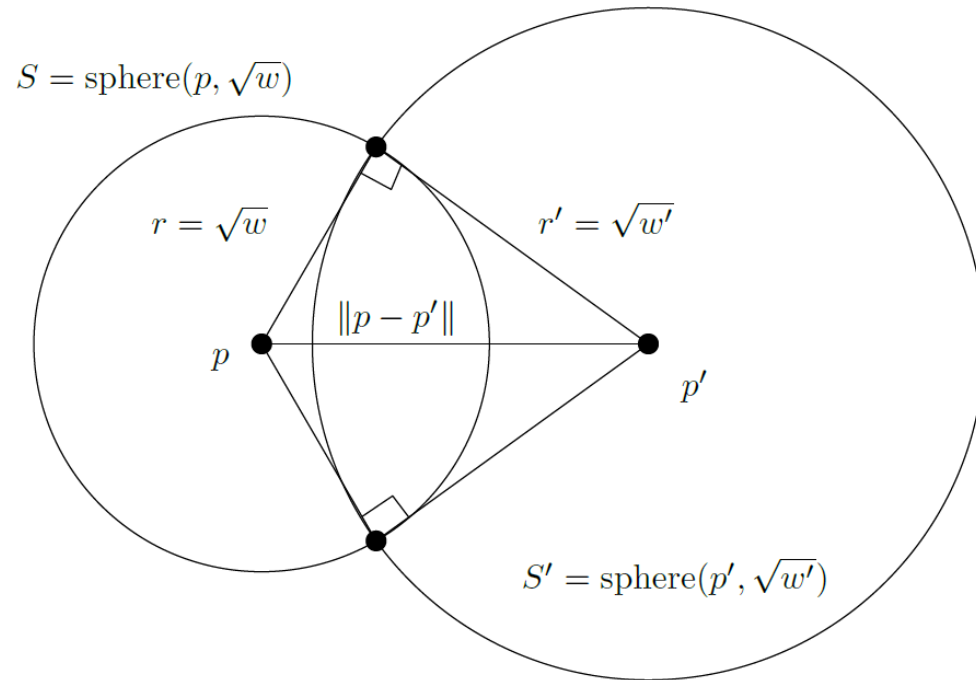
$$\hat{p}_i = \check{\eta}_i$$

$$\check{\eta}_i = (\eta_i = \nabla F(\theta_i), \|\eta_i\|^2 - 2F^*(\eta_i))$$

$$\sigma(p, \hat{p}') := \|p - p'\|^2 - w'$$

Vanishing power distance = Sphere orthogonality

$$\sigma(\hat{p}, \hat{p}') = 0 \Leftrightarrow \|p - p'\|^2 = w + w' \Leftrightarrow S \perp S'$$



$$\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$$

Pythagoras' theorem:

Power distance vanishes when spheres are orthogonal

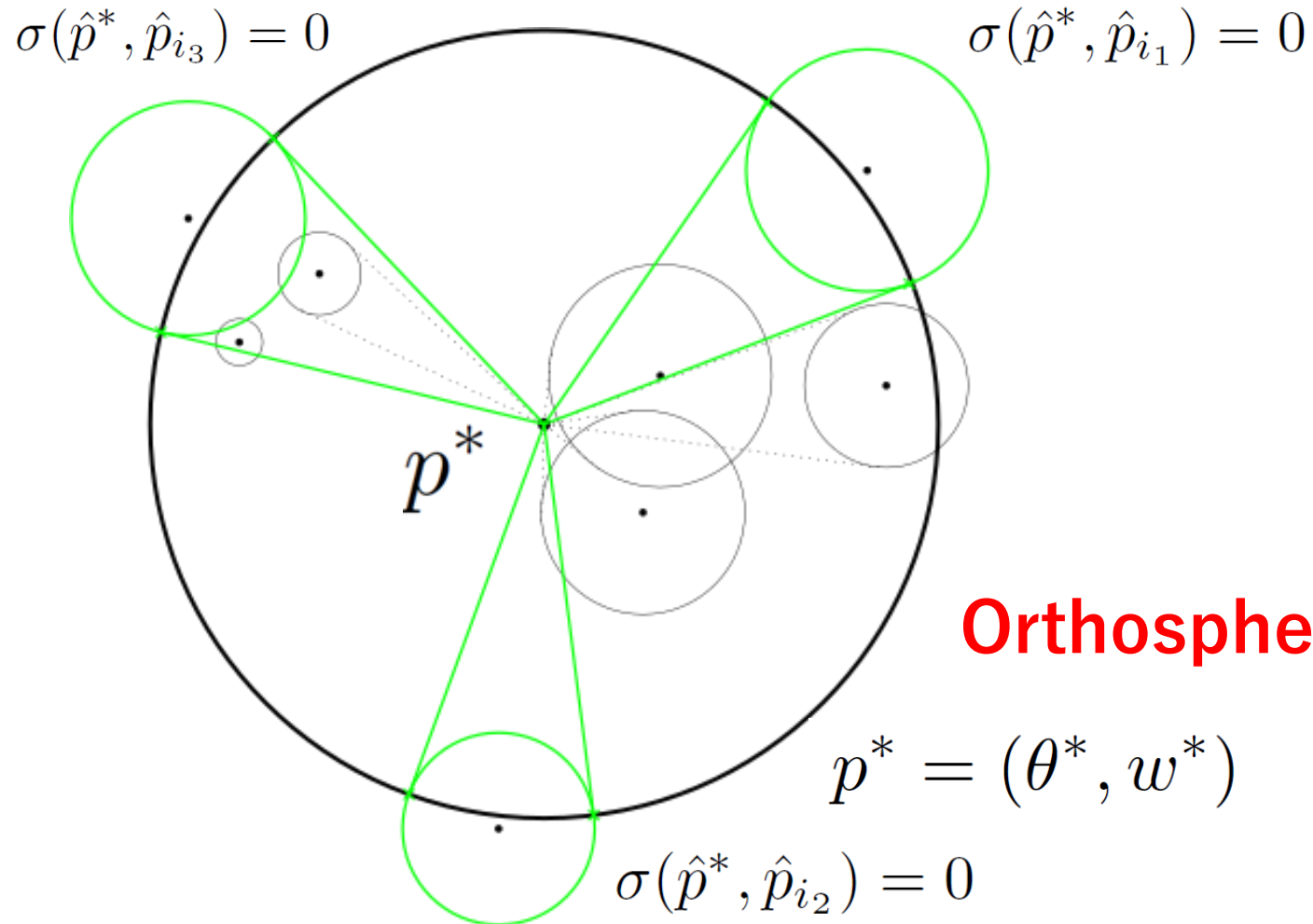
Power circumcenter and support points on the MEB boundary:

$$\sigma(\theta^*, \hat{p}_i) = w^* \Rightarrow \sigma(\hat{p}^*, \hat{p}_i) = 0 \quad \hat{p}^* = (\theta^*, w^*)$$

$S^* \perp S_i$ **Support spheres are orthogonal to power MEB**

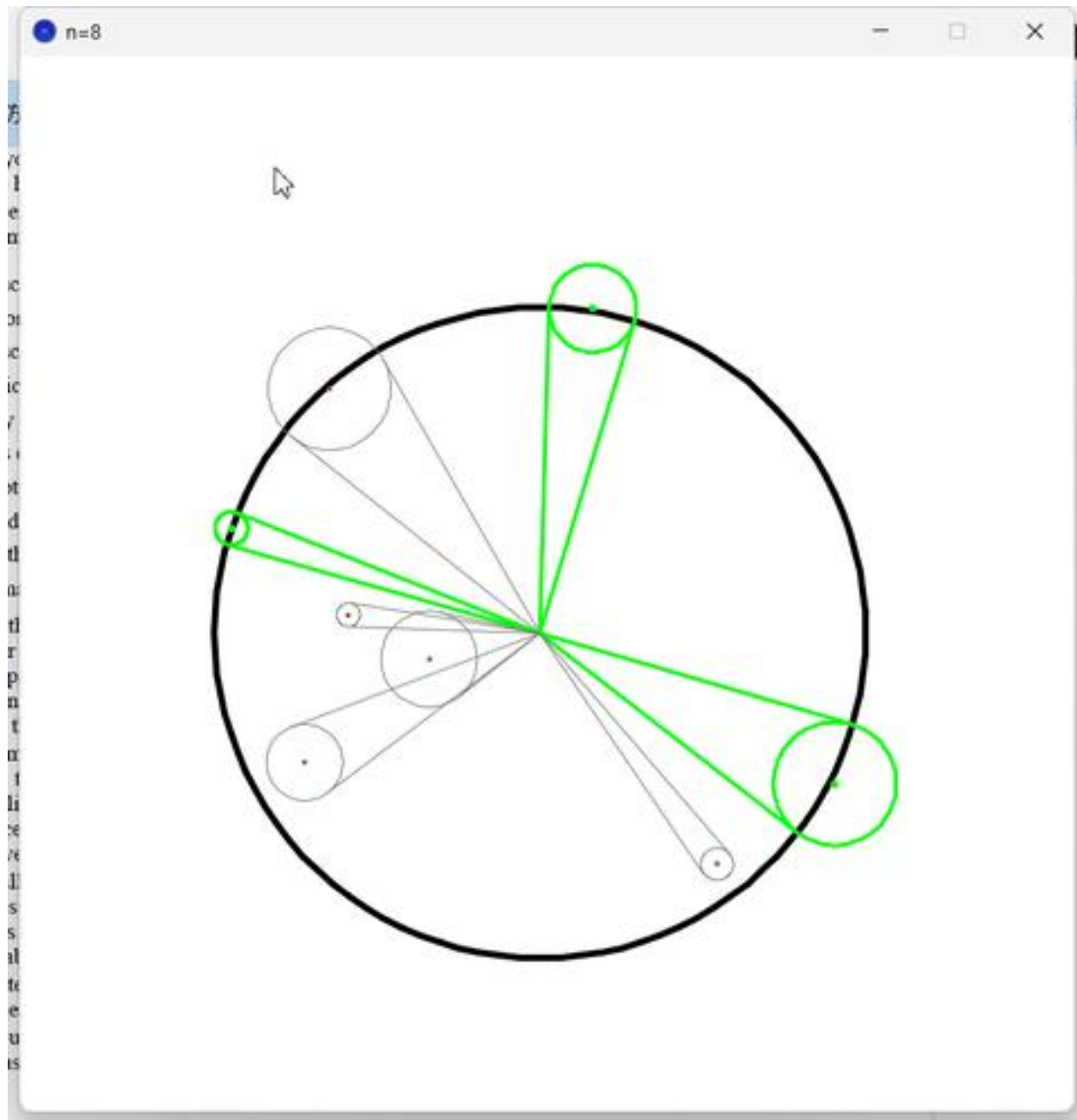
Bregman MEBs as power MEBs

Use computational geometry toolbox in information geometry!



Bregman miniball in θ
as equivalent
power miniball in η

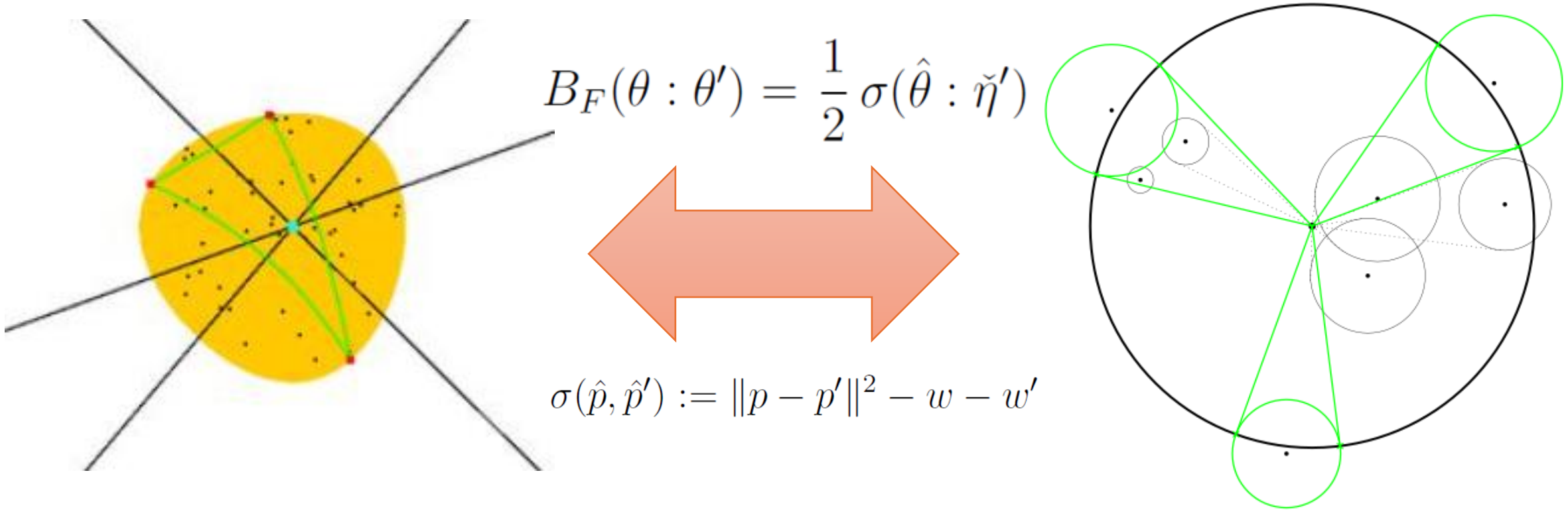
Orthosphere of support spheres



- Once Bregman parameters are converted to gradient parameters, consider only weighted squared quadratic distance to design algorithms: No need to remind Bregman generator
- Useful for analysis proof of convergences, etc.

Bregman divergences as power distances: CG

Use toolbox of computational geometry in information geometry!



Bregman miniball in θ

as Power miniball in η

Right viewpoint: Do not care of generator F

Frank-Wolfe coreset power MEB approximations

$$\arg \max_{i \in [n]} B_{F^*}(\eta_i : c_t) = \arg \max \sigma(c_t, \hat{q}_i) \quad \hat{q}_i = (\eta_i, 2\omega_{F^*}(\eta_i) = \|\eta_i\|^2 - 2F^*(\eta_i))$$

Algorithm 6: FWPOWERMEB($\hat{\mathcal{S}} = \{\hat{p}_i = (p_i, w_i)\}, T$)

Choose at random $c_1 \in \{\xi_i\}$;

for $t = 1, 2, \dots, T - 1$ **do**

$f_t \leftarrow \arg \max_{i \in [n]} \sigma(c_t, \hat{p}_i)$; /* farthest power distance input point */
 $c_{t+1} = \frac{t}{t+1}c_t + \frac{1}{t+1}p_{f_t}$; /* circumcenter walks into farthest point direction */

end

Return $\tilde{c} = c_T$;

- 1/ Frank-Wolfe analysis and variants
- 2/ ... can be kernelized

Frank-Wolfe: Bregman MEB vs power MEB

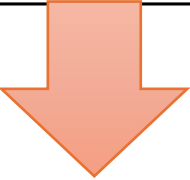
Algorithm 4: Left Bregman MEB approximation algorithm $\text{BREGMANBC}(\mathcal{T} = \{\theta_i\}, T)$

Choose at random $c_1 \in \mathcal{T}$;
for $t = 1, 2, \dots, T - 1$ **do**
 $f_t \leftarrow \arg \max_{i \in [n]} B_F(c_t : \theta_i)$;
 $c_{t+1} \leftarrow (\nabla F)^{-1} \left(\frac{t}{t+1} \nabla F(c_t) + \frac{1}{t+1} \nabla F(\theta_{f_t}) \right)$;
end
Return $\tilde{c} = c_T$;

θ -view

[Nock & N, 2005]

No proof of approximations

Gradient map  $\eta = \nabla F(\theta)$

Algorithm 5: $\text{BREGMANBCETA}(\mathcal{E} = \{\eta_i\}, T)$

Choose at random $c_1 \in \mathcal{E}$;
for $t = 1, 2, \dots, T - 1$ **do**
 $f_t \leftarrow \arg \max_{i \in [n]} B_{F^*}(\eta_i : c_t)$;
 $c_{t+1} \leftarrow \frac{t}{t+1} c_t + \frac{1}{t+1} \eta_{f_t}$;
end
Return $\tilde{c} = c_T$;

η -view

Algorithm 6: $\text{FWPOWERMEB}(\hat{\mathcal{S}} = \{\hat{p}_i = (p_i, w_i)\}, T)$

Choose at random $c_1 \in \{\xi_i\}$;
for $t = 1, 2, \dots, T - 1$ **do**
 $f_t \leftarrow \arg \max_{i \in [n]} \sigma(c_t, \hat{p}_i)$;
 $c_{t+1} = \frac{t}{t+1} c_t + \frac{1}{t+1} p_{f_t}$;
end
Return $\tilde{c} = c_T$;

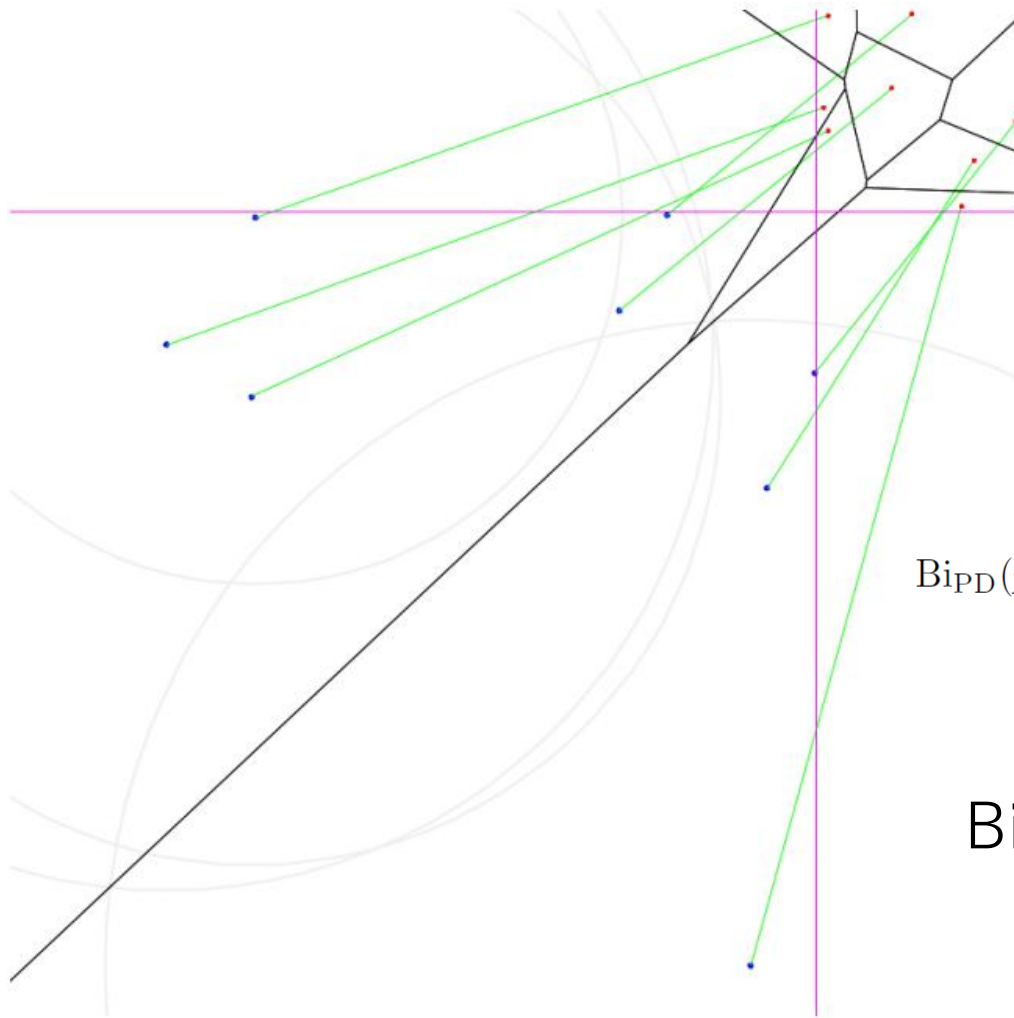
Weighted η -point view

Power distance view

With **proof** of approximations

Bregman Voronoi diagrams as power diagrams

Bregman Voronoi in θ as **clipped power diagrams** in η



$$\begin{aligned}\text{Bi}_F(\theta_1, \theta_2) &:= \{\theta \in \Theta : B_F(\theta : \theta_1) = B_F(\theta : \theta_2)\} \\ &= \left\{ \theta \in \Theta : \frac{1}{2}\sigma(\hat{\theta} : \check{\eta}_1) = \frac{1}{2}\sigma(\hat{\theta} : \check{\eta}_2) \right\} \\ &= \{\theta \in \mathbb{R}^m : \sigma(\theta : \check{\eta}_1) = \sigma(\theta : \check{\eta}_2)\} := \text{Bi}_{\text{PD}}(\check{\eta}_1, \check{\eta}_2) \cap \Theta\end{aligned}$$

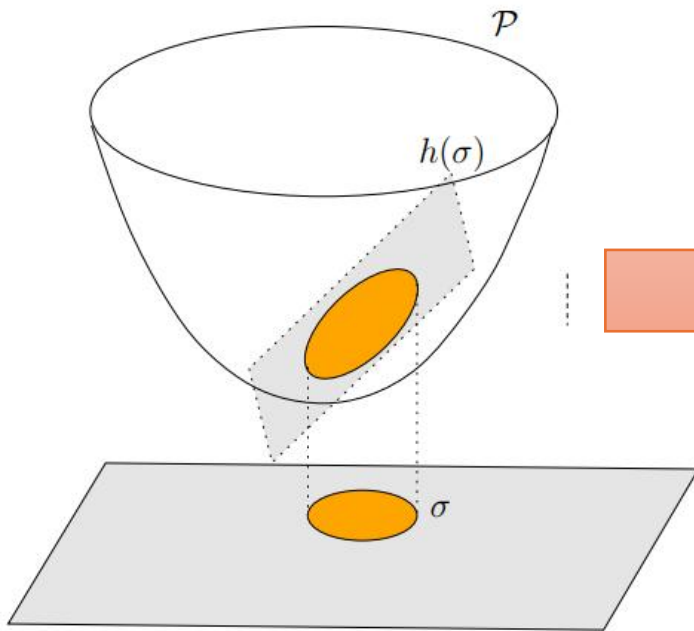
$$\check{\eta}_i = (\eta_i = \nabla F(\theta_i), \|\eta_i\|^2 - 2F^*(\eta_i))$$

$$\begin{aligned}\text{Bi}_{\text{PD}}(\hat{p}_1 = (p_1, w_1), \hat{p}_2 = (p_2, w_2)) &= \{p \in \mathbb{R}^m : \sigma(p, \hat{p}_1) = \sigma(p, \hat{p}_2)\} \\ &= \{p \in \mathbb{R}^m : \|p - p_1\|^2 - w_1 = \|p - p_2\|^2 - w_2\} \\ &= \{p \in \mathbb{R}^m : 2\langle p, p_2 - p_1 \rangle + \|p_1\|^2 - \|p_2\|^2 - w_1 + w_2 = 0\}\end{aligned}$$

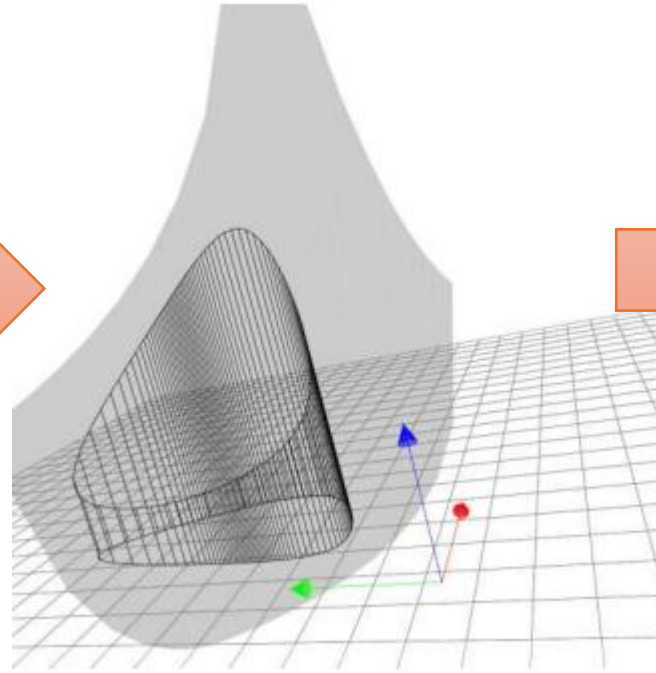
Bisectors are hyperplanes: affine Voronoi diagrams

Direct derivation from BVD-PD equivalence

Space of Bregman spheres & hyperplane remapping

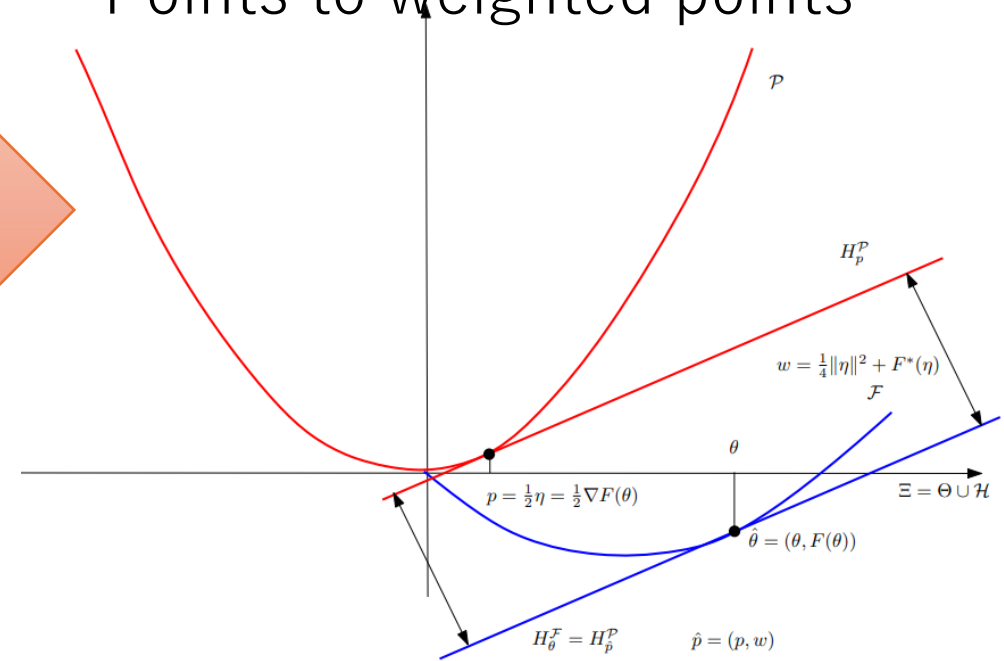


Euclidean sphere
from paraboloid lifting



Bregman sphere
from potential lifting

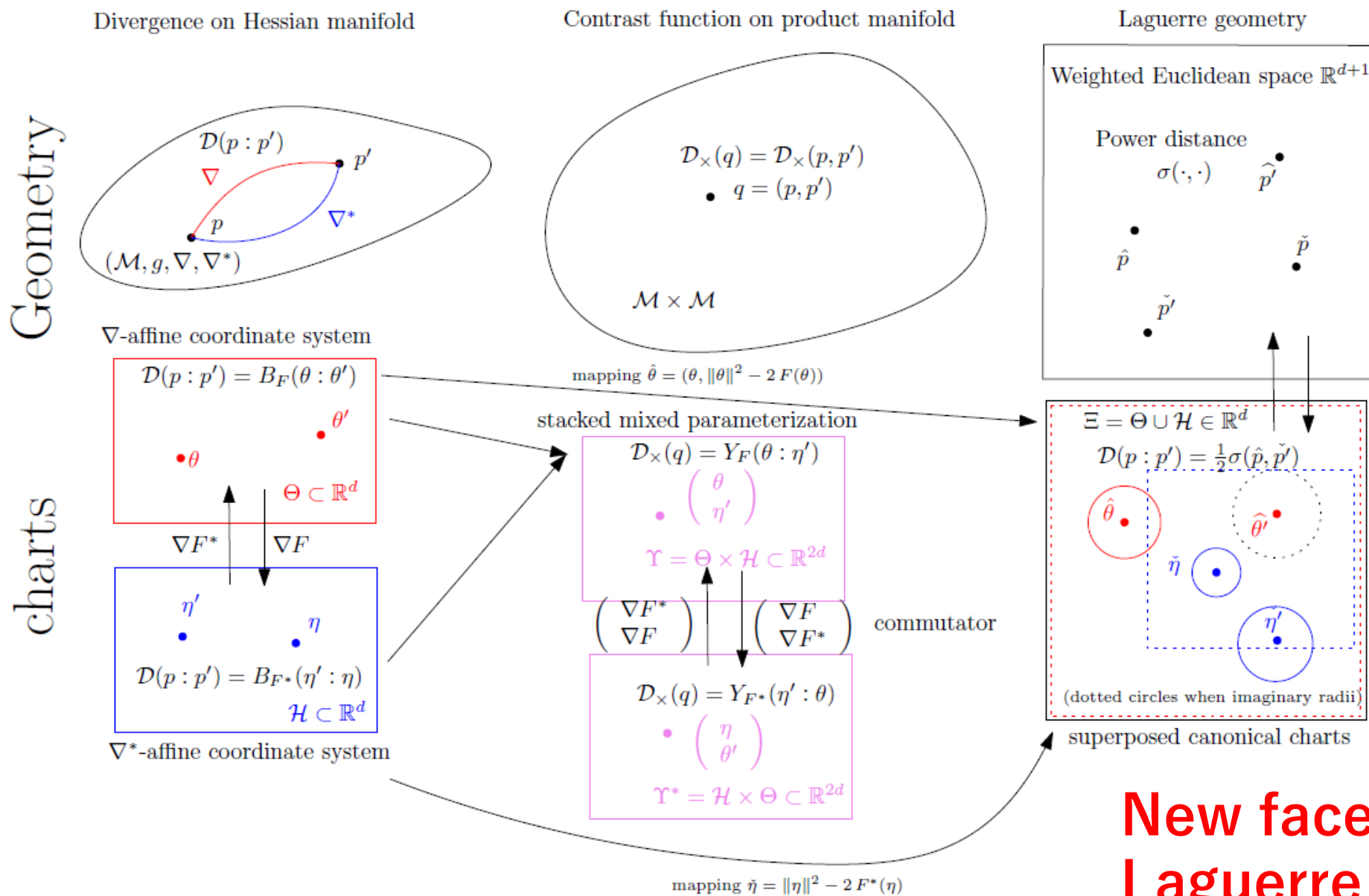
Remap tangent planes between
two potential functions:
Points to weighted points



$$\begin{aligned}\hat{p} &= \left(\frac{1}{2}\eta, \frac{1}{4}\|\eta\|^2 + \langle \theta, \eta \rangle - F(\theta) \right) \\ &= \left(\frac{1}{2}\eta, \frac{1}{4}\|\eta\|^2 + F^*(\eta) \right).\end{aligned}$$

Boolean algebra of halfspaces...

Geometric interpretation: superposed charts



**New face of BD:
Laguerre geometry**

Convex conjugates from quadratic polarities*

*Joint work with Basile Plus-Gourdon and Mahito Sugiyama

Polarity of a **set** A : $\Delta(A) := \{[b] \in \mathbb{R}_{n+1} \quad : \quad \forall [a] \in A, \quad p(a, b) \geq 0\}$

Quadratic polarity when $p(a, b) := [a]^\top C [b]$

Property (Legendre transformation from Legendre polarity). *Let $F : \Theta \rightarrow \mathbb{R}$ be a real-valued function and $F^* : H \rightarrow \mathbb{R}$ be its Legendre convex conjugate. Then it holds that*

[Fenchel 1949]

$$\partial \Delta_{\mathcal{L}}(\text{graph}(F)) = \text{graph}(F^*)$$

Legendre polarity $C = C_{\mathcal{L}} := \begin{bmatrix} -I_n & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

n-variate function $\rightarrow$ Graph $(n+1)$ -dim $\rightarrow$ Projective space/homogeneous coord. **(n+2)-dim**

Fenchel-Young divergences as inner products

F a Legendre-type strictly convex function with convex conjugate F^* , $(F^*)^*=F$
Fenchel-Young divergences are equivalent to Bregman divergences:

$$Y_F(\theta_1 : \eta_2) := F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle = Y_{F^*}(\eta_2 : \theta_1) \geq 0$$

$$= \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle$$

GPU!

Geometric interpretation: Legendre transform considered as a **polarity**

$$Y_F(\theta_1 : \eta_2) = (\tilde{\theta}_1^\uparrow)^\top C_{\mathcal{L}} \tilde{\eta}_2^\uparrow, \quad C_{\mathcal{L}} := \begin{bmatrix} -I & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \longrightarrow \quad Y_F(\theta_1 : \eta_2) = \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle$$

where $\tilde{\theta}^\uparrow := \begin{bmatrix} \theta \\ F(\theta) \\ 1 \end{bmatrix}$, $\tilde{\eta}^\uparrow := \begin{bmatrix} \eta \\ F^*(\eta) \\ 1 \end{bmatrix}$ are homogeneous vectors of points on the graphs of potential functions

Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv: 2606.09077

Legendre transform and gauge transformations*

*Joint work with Basile Plus-Gourdon and Mahito Sugiyama

Quadratic Polarity realized as Legendre Polarity $\rightarrow$ Deformation

Theorem 1 (Quadratic polarity as transformed convex body of the Legendre polarity). *Let $C \in \text{GL}(n+2)$ and suppose there exists $\mathcal{M}_T \in \text{GL}(n+2)$ such that $C = C_{\mathcal{L}} \mathcal{M}_T^{-1}$, where T is an affine deformation defined by its matrix $\mathcal{M}_T = C^{-1} C_{\mathcal{L}}$ as $T : [b] \mapsto \mathcal{M}_T[b]$. Then for any $A \subset \mathbb{R}^{n+1}$, we have $\Delta_C(A) = T(\Delta_{\mathcal{L}}(A))$.*

Quadratic Polarity realized as Deformation $\rightarrow$ Legendre Polarity

Theorem 2 (Quadratic polarity as Legendre polarity on deformed convex body). *Let $C \in \text{GL}(n+2)$ and suppose there exists $\mathcal{M}_S \in \text{GL}(n+2)$ such that: $C = \mathcal{M}_S^T C_{\mathcal{L}}$, where S is an affine deformation defined by its matrix $\mathcal{M}_S = C_{\mathcal{L}} C^T$ as $S : [a] \mapsto \mathcal{M}_S[a]$. Then for any $A \subset \mathbb{R}^{n+1}$: $\Delta_C(A) = \Delta_{\mathcal{L}}(S(A))$.*

with correspondences: $\mathcal{M}_T = C_{\mathcal{L}} \mathcal{M}_S^{-T} C_{\mathcal{L}}$  $\mathcal{M}_S = C_{\mathcal{L}} \mathcal{M}_T^{-T} C_{\mathcal{L}}$

Gauge freedom with matrix pipeline

Convex conjugates F^* via neural networks*

*Joint work with Basile Plus-Gourdon

2606.09077

- Conjugate $F^*(\eta) = \langle (\nabla F)^{-1}(\eta), \eta \rangle - F((\nabla F)^{-1}(\eta))$

may not be in closed form when $(\nabla F)^{-1}$ not available or computationally intractable: Often the case in ML and optimal transport.

- Conjugation map:
$$\theta^*(\eta) = \operatorname{argmax}_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - f(\theta) \} = \nabla f^*(\eta)$$

parameterize a *residual network* $\theta_\varphi(\eta) = \eta + \text{MLP}_\varphi(\eta)$, with the multilayer perceptron (MLP)

Motivation: Half quadratic paraboloid yield identity, linear gradient map

- Train NN with loss function : $\mathcal{L}(\varphi) = \mathbb{E}_\eta [f(\theta_\varphi(\eta)) - \langle \theta_\varphi(\eta), \eta \rangle]$
- Once trained, we *approximate the convex conjugate* by

$$\hat{f}^*(\eta) = \langle \theta_\varphi(\eta), \eta \rangle - f(\theta_\varphi(\eta))$$

- We use **gauge freedom** to **stabilize numerically** the NN learning...

Discussion

Edmond Nicolas Laguerre



Born 9 April 1834
Bar-le-Duc, France

Died 14 August 1886 (aged 52)
Bar-le-Duc, France

Laguerre geometry is a branch of mathematics founded by French mathematician Edmond Laguerre that studies the properties of **oriented hyperplanes and oriented spheres** under Laguerre transformations.

https://en.wikipedia.org/wiki/Laguerre_transformations

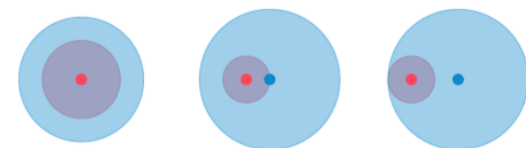
[Submitted on 12 Jul 2026]

Laguerre Geometry for Interpreting Large Language Models

Chunwei Ma, Russell Wolfinger

(A) Oriented spheres in Laguerre Geometry

concentric subsumption internal tangent



intersecting external tangent disjoint



https://horsepurve.github.io/interpret-llm/posts/01_llm_lens/

Functional Bregman divergences

Definition 6 (Implied by Exercise 17.8 of [Bauschke and Combettes \(2020\)](#)) *Let $\Phi : \mathcal{F} \rightarrow (-\infty, \infty]$ be a generator (see Definition 1) on $\mathcal{F}$. The functional Bregman divergence (FBD) $d_\Phi : \mathcal{A} \times \mathcal{A} \rightarrow (-\infty, \infty)$ generated by Φ is defined by*

$$d_\Phi(f, g) = \Phi(f) - \Phi(g) - \langle \nabla \Phi(g), f - g \rangle_{\mathcal{F}},$$

where $\nabla \Phi(g)$ is the Fréchet gradient of Φ at g .

Proposition 7 *The FBD d_Φ in Definition 6 satisfies the following basic properties:*

- 1. For all $f, g \in \mathcal{A}$, $d_\Phi[f, g] \geq 0$ and $d_\Phi[f, g] = 0$ if and only if $f = g$.*
- 2. For fixed q , the function $d_\Phi(\cdot, q) : \mathcal{F} \rightarrow [0, \infty)$ is strictly convex.*
- 3. For any two generators (see Definition 6) $\Phi_1 : \mathcal{F} \rightarrow \mathbb{R}$ and $\Phi_2 : \mathcal{F} \rightarrow \mathbb{R}$ and $\lambda \geq 0$, we have that $d_{\Phi_1 + \lambda \Phi_2} = d_{\Phi_1} + \lambda d_{\Phi_2}$.*
- 4. $d_\Phi(f, g) = d_\Phi(f, h) + d_\Phi(h, g) - \langle \nabla \Phi(g) - \nabla \Phi(h), f - h \rangle_{\mathcal{F}}$.*

Maximum mean discrepancy

$$\text{MMD}_k(f, g) = \sup_{\substack{h \in \mathcal{H} \\ \|h\|_{\mathcal{H}} \leq 1}} \mathbb{E} h(x) - h(y) = \|\mu(f) - \mu(g)\|_{\mathcal{H}} = \sqrt{\mathbb{E} k(x, x') - 2k(x, y) + k(y, y')},$$

Deformed MMDs

Definition 16 *Let $\phi : [0, \infty) \rightarrow \mathbb{R}$ be twice continuously Fréchet differentiable on $(0, \infty)$, such that $\phi'(r) > 0$ for $r > 0$ and $\phi''(r) > 0$ for $r > 0$. A functional $\Phi : \mathcal{F} \rightarrow \mathbb{R}$ of the form $\Phi(f) = \phi(\|\mu(f)\|_{\mathcal{H}})$ is called a deformed generator.*

$$d_{\Phi}(f, g) = \phi(\|\mu(f)\|_{\mathcal{H}}) - \phi(\|\mu(g)\|_{\mathcal{H}}) - \frac{\phi'(\|\mu(g)\|_{\mathcal{H}})}{\|\mu(g)\|_{\mathcal{H}}} \langle \mu(g), \mu(f) - \mu(g) \rangle_{\mathcal{H}}$$

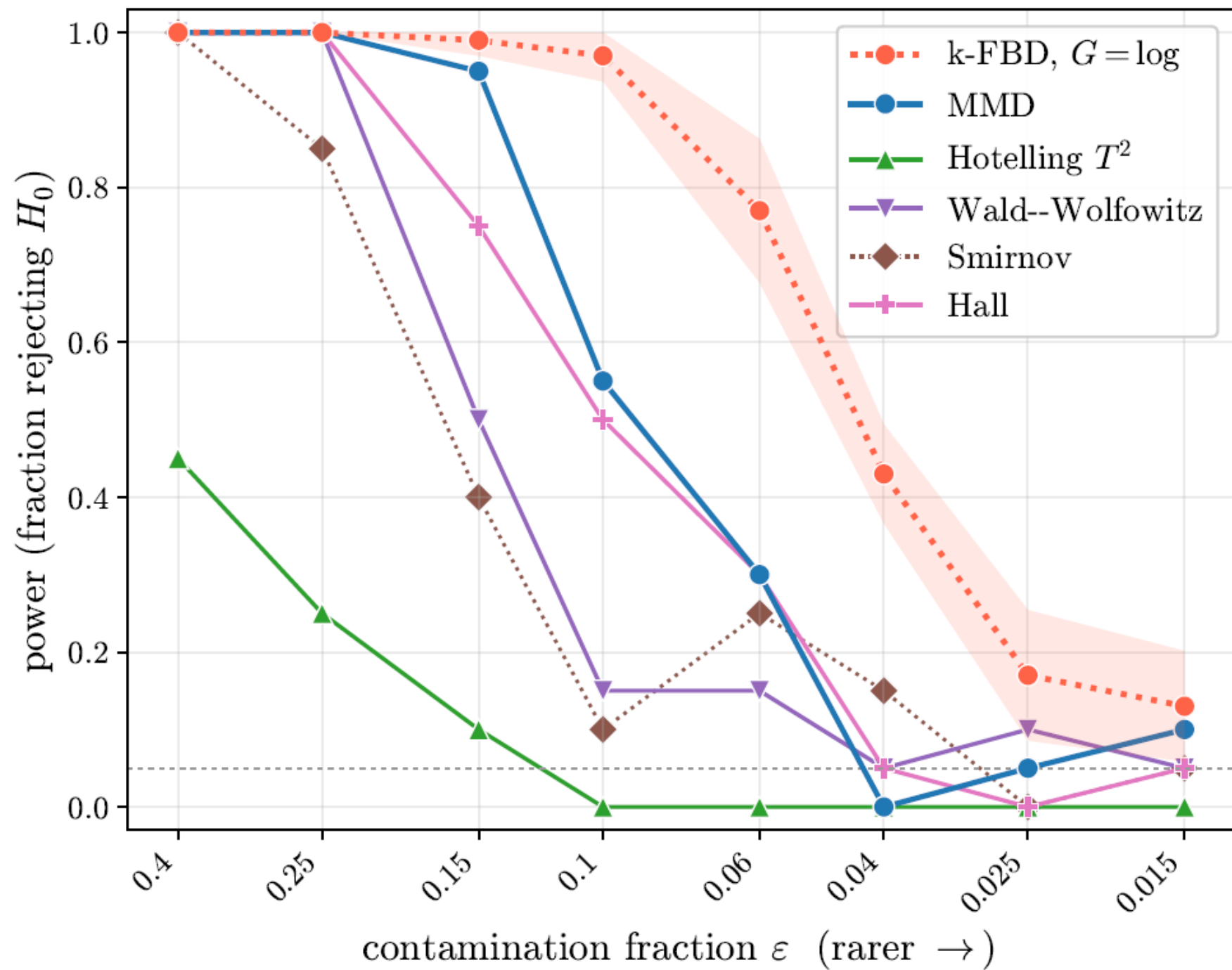
Kernelized functional Bregman divergences

Proposition 12 *Consider a Bregman functional $\Phi : \mathcal{F} \rightarrow \mathbb{R}$ of the form $\Phi(f) = F(\mu(f))$, where $\mu(f)$ is the KME of f , and $F : \mathcal{H} \rightarrow \mathcal{R}$ is a generator. Then for all $f, g \in \mathcal{A}$,*

$$d_{\Phi}(f, g) = d_F(\mu(f), \mu(g)).$$

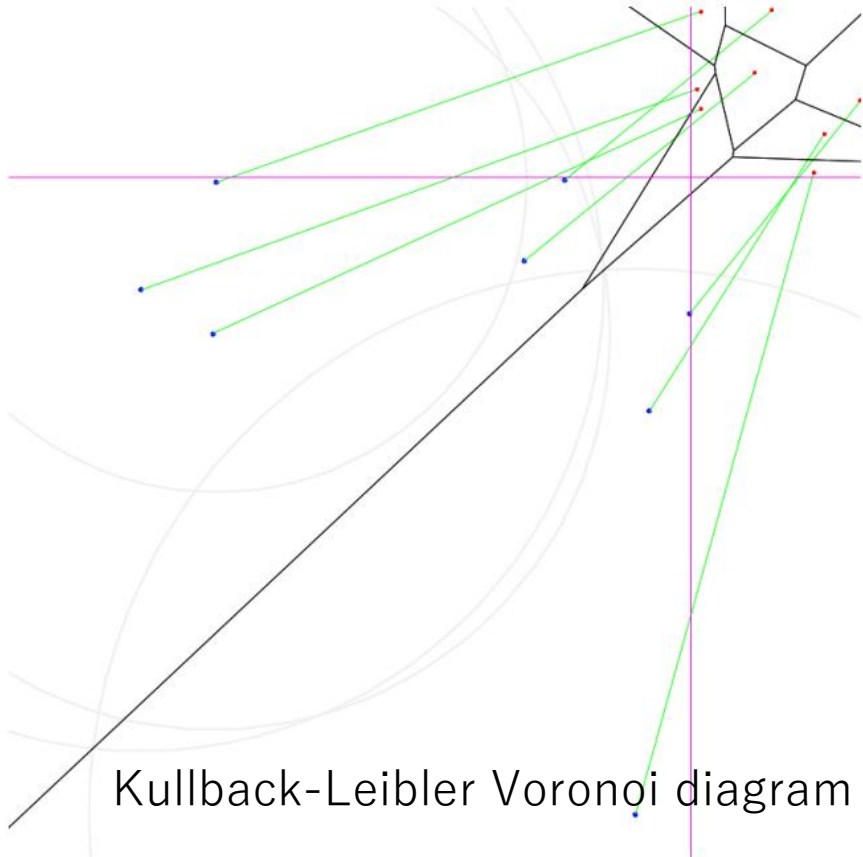
Moreover, suppose μ is injective on $\mathcal{A}$. Then $\Phi = F \circ \mu$ is a generator (as in Definition 1).

The squared MMD is obtained when $F(\cdot) = (\cdot)^2$.

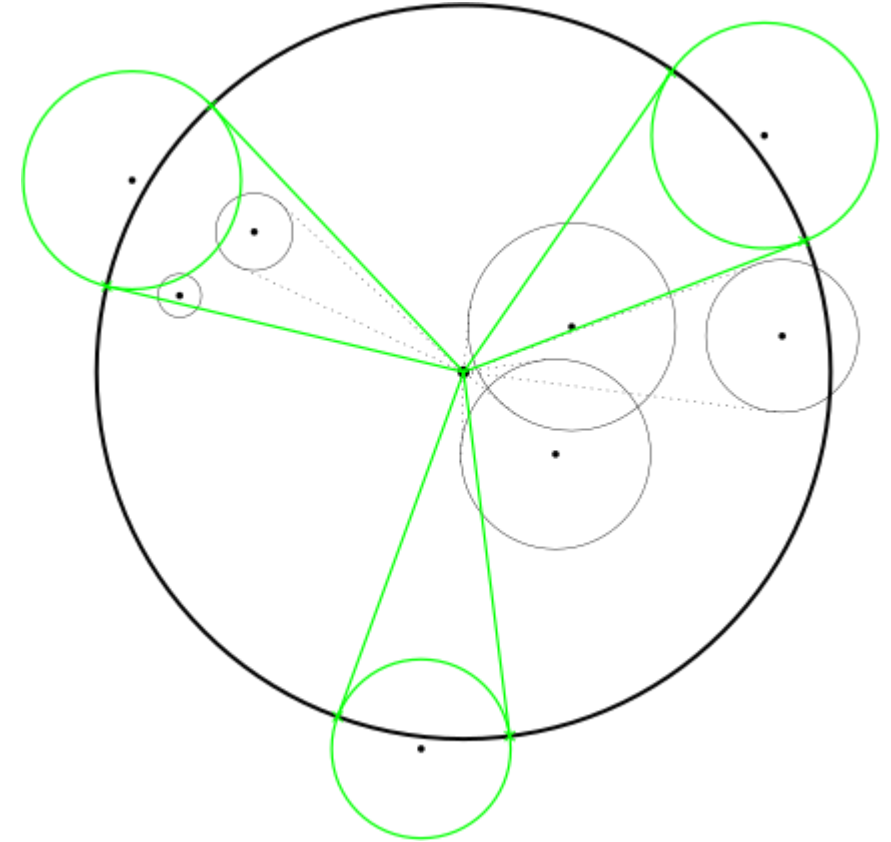


Bregman divergences as power distances

Use computational geometry toolbox in information geometry!



Bregman Voronoi in θ as
clipped power diagrams in η



Bregman miniball in θ as
Power miniball in η [2607.24197](#)

Convex duality via Legendre-Fenchel transform

- Legendre-Fenchel transform of a convex function F :

$$F^*(\eta) = \sup_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - F(\theta) \} = \sup_{\theta \in \Theta} \{ H_{\theta}(\eta) \}$$

Supremum of affine functions. Involution $(F^*)^* = F$

Graph(F): pointwise supremum of all affine functions majorized by F

- Convex conjugate: $F^*(\eta) = \langle (\nabla F)^{-1}(\eta), \eta \rangle - F((\nabla F)^{-1}(\eta))$
since sup obtained when $\nabla F(\theta) = \eta$.
- Dual Bregman divergences/swapping param.: $B_F(\theta_1 : \theta_2) = B_{F^*}(\eta_2 : \eta_1)$

Dual Fenchel-Young/Bregman divergences

- Young inequality: $F(\theta_1) + F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$
with equality iff. $\eta_2 = \nabla F(\theta_1)$
- Build the **Fenchel-Young divergence** from this inequality: lhs-rhs ≥ 0
 $Y_{F, F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$
- Mixed and dual parameterizations of an underlying geometric divergence
 $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$

$$B_F(\theta_1 : \theta_2) = Y_{F, F^*}(\theta_1 : \eta_2) = Y_{F^*, F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Geometrically, it corresponds to the expression of a point divergence on a Hessian manifold expressed in the θ -chart and the η -chart